

The Relationship Between CO₂ Emissions and Global Temperature Anomalies

Jacob Pauley

May 1, 2025

Abstract

This study investigates the relationship between global carbon dioxide (CO₂) emissions and global surface temperature anomalies. Using publicly available data from NASA and Our World in Data, we merged annual CO₂ emissions (in million tonnes) with global temperature anomaly data from 1880 to the present. A Pearson correlation and linear regression analysis were performed. Results indicate a strong positive relationship between CO₂ emissions and temperature anomalies, supporting the hypothesis that increasing greenhouse gas emissions contribute to global warming. This finding highlights the urgency of emissions mitigation in global climate policy.

Introduction

The impact of human activity on Earth's climate has become a critical area of scientific inquiry. Among the most significant contributors to global climate change are greenhouse gas emissions, particularly carbon dioxide (CO₂). This study aims to evaluate the relationship between historical global CO₂ emissions and global temperature anomalies. We hypothesize that increasing CO₂ emissions are positively correlated with increases in global temperature anomalies.

Methods

Data Sources

- CO₂ Emissions: Sourced from Our World in Data's global emissions dataset.
 - Temperature Anomalies: Obtained from NASA's GISTEMP surface temperature dataset.
- The data were merged by year and analyzed using Python libraries: Pandas, Seaborn, and Scikit-learn.

Data Cleaning

- CO₂ data was filtered to include only global values.
- Monthly temperature anomalies were converted to floats and averaged to compute an annual anomaly.
- Missing values were dropped from both datasets.

Statistical Analysis

- A Pearson correlation coefficient was computed.
- A simple linear regression model was fit with CO₂ emissions as the independent variable and temperature anomaly as the dependent variable.

Results

Null Hypothesis (H_0): *There is no linear relationship between CO₂ emissions and global temperature anomalies. Mathematically: $\beta_1 = 0$ (the slope is zero)*

Alternative Hypothesis (H_1): *There is a linear relationship between CO₂ emissions and global temperature anomalies. Mathematically: $\beta_1 \neq 0$*

The Pearson correlation coefficient between global CO₂ emissions and global temperature anomalies was $r = 0.95$, indicating a strong positive relationship.

A linear regression analysis revealed that CO₂ emissions significantly predicted temperature anomalies,
 $b = 0.00003$, $t(142) = 31.05$, $p < .001$, $R^2 = .87$.

The regression equation was:

Temp_anomaly = $0.0006 * CO_2_emissions$ + Intercept

A scatter plot with a fitted regression line visually confirmed the strength and direction of the relationship.

Discussion

The results of this study support the hypothesis that rising global CO₂ emissions are strongly associated with increases in global temperature anomalies. The high correlation and model fit suggest a clear linear trend, although further studies might explore more complex or nonlinear relationships.

Limitations

- This model does not account for other greenhouse gases or natural climate drivers.
- The lag effect between emissions and climate response is not analyzed.

Future Research

Future studies could examine the role of methane, land use, and regional temperature responses, or use time-series models to predict future warming.

References

NASA Goddard Institute for Space Studies. (n.d.). GISS Surface Temperature Analysis (GISTEMP). <https://data.giss.nasa.gov/gistemp/>

Ritchie, H., Roser, M., & Rosado, P. (2020). CO₂ and Greenhouse Gas Emissions. Our World in Data. <https://ourworldindata.org/co2-and-other-greenhouse-gas-emissions>